

Composed and Compiled by Imtiaz Albaloshi:

Bt302-Immunology:

Midterm objective:

Whatsapp: 03196386297

QNo-1:

1. The collection of cells, tissues and molecules that mediate resistance to infection is called the immune system.
2. These are contagious, transmissible and communicable diseases which are caused by an invading or infectious agent.
3. The infectious agent is a foreign particle to the body's immune system.
4. Typhoid is a bacterial infection while hepatitis is a viral infection.
5. Infections can be transmitted from one individual to another through various ways like aerosols, blood and also from mother to fetus.
6. These are non-communicable and non-contagious diseases in contrast to infections.
7. Non-infectious diseases are generally considered as metabolic disorders.
8. The most important function of the immune system is to protect the body from the attack of infections or invading microbes.

9. Innate immune system is non-specific.

10. The adaptive immune system is considered as the second line of protection against various pathogens specifically.

11. For the activation of the adaptive immune system, it needs lag time for full activation during which naive cells become active after differentiation and activation.

12. Adaptive immune system has the unique attribute of memory against the specific nature of pathogens.

13: Innate immune system is the first line of defense against invading pathogens.

14. The peristalsis and ciliary movements of GIT also protect the ingested microorganisms.

15. Flushing actions of tears and saliva also make the mechanical arm of anatomical barriers.

16: Lungs also contain surfactants which increase the process of phagocytosis for potential pathogens by the process of opsonization.

17: Similarly, tears and nasal secretions carry various enzymes like lysozymes and phospholipases for degradation of invading pathogens by breaking their cell wall.

18: Biological arms consist of normal flora or microbiota of various structures of the body like skin, GIT, respiratory tract and genital tract.

19: Humoral barriers in fact mediate that inflammatory process.

20. Complement systems are plasma proteins which are present in inactive form.

21: For example, beta-lysin, a protein produced by platelets during coagulation, can lyse many Gram positive bacteria by acting as a cationic detergent.

22: Lactoferrin and transferrins are the proteins which limit bacterial growth.

23: Interferons are the secretory molecules which are proteins in nature.

24: PMNs contribute to collateral tissue damage that occurs during inflammation.

25: NK cells can nonspecifically kill virus infected and tumor cells.

26: Neutrophils are motile phagocytic cells that have lobed nuclei.

27: Macrophages are phagocytic cells that have a characteristic kidney-shaped nucleus.

28: Some of the SOS signals stimulate endothelial cells near the site of the infection to express cell adhesion molecules such as intracellular Adhesion Molecules-1.

29: Scavenger receptors bind a wide variety of polyanions on bacterial surfaces resulting in phagocytosis of bacteria.

30: Phagocytes have a variety of Toll-like receptors which recognize broad molecular patterns called PAMPs.

31: After attachment of a bacterium, the phagocyte begins to extend pseudopods around the bacterium.

32: During phagocytosis the granules or lysosomes of the phagocyte fuse with the phagosome and empty their contents.

33: During phagocytosis there is an increase in glucose and oxygen consumption which is referred to as the respiratory burst.

34: As the azurophilic granules fuse with the phagosome, myeloperoxidase is released into the phagolysosome.

35: Myeloperoxidase utilizes H₂O₂ and halide ions to produce hypochlorite, a highly toxic substance.

36: Lectin pathway is considered very similar to classical pathway in context to activation of complement protein 1.

37: C3 proteins are produced after interaction of mannose and lectins.

38: Anaphylaxis are also known as Anaphylatoxins.

39: Complement proteins are considered as a major humoral component of the innate immune system.

40: Secondary lymphoid tissues include lymph nodes in various parts of the body like axillary and groin regions.

41: Spleen is located in the left hypochondriac region.

42: Appendix and peyers patches are located in the gastrointestinal tract.

43: Myeloid lineage of immune cells from hematopoietic cells give rise to a variety of immune cells like monocytes and macrophages.

44: Naive lymphocytes enter the lymph nodes from the blood via High Endothelial Venules (HEVs).

45: Homing receptors on the lymphocytes direct the cells to the HEVs.

46: It is estimated that about 1-2% of lymphocytes recirculate every hour.

47: Immune system provides protection against diverse pathogens which are highly contagious in nature.

48: Antibody binding to a pathogen can opsonize the material and facilitate its uptake and destruction by phagocytic cells.

49: If there is a second exposure of the body with the same immunogen, it is called a secondary injection of immunogen.

50: Equilibrium phase is the rapid diffusion process.